

Olympiad Test Paper — Mathematics (Class 7) — Paper 4

Chapter 7: A Tale of Three Intersecting Lines

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	7 — A Tale of Three Intersecting Lines
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. An exterior angle equals the SUM of the two remote interior angles (never the adjacent one); a sum of two sides must strictly EXCEED the third; 'acute' needs all three angles below 90° ; and the largest angle always sits opposite the longest side. Do not write on this paper — mark answers on your answer sheet.

1. In triangle ABC the angles are in the ratio 4 : 5 : 6. Side AB is extended beyond A, forming an exterior angle at A. What is the measure of that exterior angle?

- (A) 48° (B) 108°
(C) 132° (D) 72°

2. Two sides of a triangle measure 17 cm and 8 cm and the third side is a whole number of centimetres. How many different whole-number values can the third side take?

- (A) 17 (B) 16
(C) 9 (D) 15

3. In triangle ABC, $\angle A = 3x + 5^\circ$, $\angle B = 2x - 5^\circ$ and $\angle C = 4x$. The largest angle of the triangle measures:

- (A) 65° (B) 35°
(C) 85° (D) 80°

4. In triangle PQR, side PR is extended beyond R to S so that $\angle QRS = 132^\circ$. Inside the triangle, $\angle P$ is twice $\angle Q$. Find $\angle P$.

- (A) 44° (B) 92°
(C) 88° (D) 48°

5. In an isosceles triangle ABC, $AB = AC$ and the bisector of the vertex angle $\angle A$ meets BC at D. In the smaller triangle ABD, $\angle ADB = 90^\circ$ and $\angle BAD = 34^\circ$. What is the vertex angle $\angle BAC$ of the whole triangle?

- (A) 34° (B) 68°
(C) 56° (D) 112°

6. A triangle has sides of length 11 cm and 11 cm and the third side is a whole number. How many whole-number values can the third side take?

- (A) 22 (B) 21
(C) 11 (D) 20

7. In triangle ABC, $\angle A = 90^\circ$ and M is the midpoint of the hypotenuse BC, with AM drawn. Using only triangle-angle facts, which statement is guaranteed?

- (A) $\angle AMB$ is the exterior angle of triangle AMC at M, so $\angle AMB = \angle MAC + \angle MCA$
(B) AM is perpendicular to BC (C) AM bisects the right angle at A
(D) AM equals the side AB

8. In triangle ABC, $\angle B = \angle C$, and the side opposite $\angle A$ measures 14 cm. The triangle's perimeter is 50 cm. What is the length of the side opposite $\angle B$?

- (A) 14 cm (B) 36 cm
(C) 22 cm (D) 18 cm

9. In triangle ABC, the exterior angle at A is $(5y)^\circ$ and the exterior angle at B is $(4y)^\circ$. The interior angle at C is 80° . Find y.

- (A) 20° (B) 28°
(C) 31° (D) $(260/9)^\circ \approx 28.9^\circ$

10. A triangle has angles measuring 42° and 96° . Its sides are listed from shortest to longest. Which ordering is correct?

- (A) Side opposite 42° = side opposite 42° < side opposite 96°
(B) Side opposite 96° < side opposite 42° < side opposite 42°
(C) All three sides are unequal with the 96° -side shortest
(D) All three sides are equal

11. In triangle ABC, side BC is extended beyond C to D. The exterior angle $\angle ACD = 4\angle A$ and $\angle B = 36^\circ$. Find $\angle A$.

- (A) 12° (B) 9°
(C) 36° (D) 48°

12. Two sides of a triangle are 9 cm and 16 cm. The third side is a whole number that is ODD. How many such values are possible?

- (A) 7 (B) 8
(C) 9 (D) 16

13. In an isosceles triangle, the vertex angle equals the SUM of the two base angles. The vertex angle measures:

- (A) 90°
- (B) 60°
- (C) 45°
- (D) 120°

14. In triangle ABC, $\angle A - \angle B = 24^\circ$ and $\angle B - \angle C = 12^\circ$. Find the largest angle.

- (A) 56°
- (B) 80°
- (C) 72°
- (D) 84°

15. A triangle has sides 8 cm, 8 cm and 8 cm; a second triangle has angles 20° , 20° and 140° . Which single statement classifies BOTH correctly?

- (A) The first is equilateral and right; the second is isosceles and acute
- (B) The first is equilateral and acute; the second is isosceles and obtuse
- (C) Both are isosceles and acute
- (D) The first is scalene; the second is equilateral

16. In triangle ABC, the exterior angle at A is 3 times the interior angle at A. The interior angle at A measures:

- (A) 60°
- (B) 135°
- (C) 30°
- (D) 45°

17. Two sides of a triangle are 23 cm and 14 cm. The third side is a whole number. The difference between the largest and smallest possible third sides is:

- (A) 26
- (B) 28
- (C) 9
- (D) 37

18. In triangle ABC, $AB = AC$ and the exterior angle at the apex A (formed by extending side BA beyond A) measures 110° . Find each base angle.

- (A) 55°
- (B) 70°
- (C) 35°
- (D) 110°

19. In triangle ABC, $\angle A = 80^\circ$ and the bisectors of $\angle B$ and $\angle C$ meet at I. What is $\angle BIC$?

- (A) 100°
- (B) 130°
- (C) 140°
- (D) 50°

20. A triangle has sides 6 cm, 6 cm and 10 cm. Classified by both sides and angles, the BEST description is:

- (A) Isosceles and acute
- (B) Scalene and obtuse
- (C) Equilateral and obtuse
- (D) Isosceles and obtuse

21. In triangle ABC, $\angle A = (x - 10)^\circ$, $\angle B = (x + 10)^\circ$ and $\angle C = (2x)^\circ$. Classified by its angles, the triangle is:

- (A) Right-angled
- (B) Acute
- (C) Obtuse
- (D) Equiangular

22. Three lengths are $(x - 2)$ cm, x cm and $(x + 2)$ cm, where x is a whole number. The smallest value of x for which these can be the sides of a triangle is:

- (A) 3 (B) 4
(C) 6 (D) 5

23. In triangle ABC, $\angle ABC = 70^\circ$ and $\angle A = 60^\circ$. Side BC is extended beyond C to D. The exterior angle $\angle ACD$ equals:

- (A) 110° (B) 130°
(C) 50° (D) 120°

24. An isosceles triangle has each base angle equal to three times the vertex angle. The vertex angle measures:

- (A) 36° (B) $(180/7)^\circ \approx 25.7^\circ$
(C) 60° (D) $(180/5)^\circ = 36^\circ$

25. Two sides of a triangle are 12 cm and 12 cm. The third side is a whole number that makes the triangle obtuse at the vertex opposite the third side. The largest such third side is:

- (A) 16 cm (B) 17 cm
(C) 24 cm (D) 23 cm

26. In triangle ABC, $\angle C = 90^\circ$. The bisectors of $\angle A$ and $\angle B$ meet at I. What is $\angle AIB$?

- (A) 90° (B) 45°
(C) 135° (D) 120°

27. In triangle ABC, $\angle A = 3\angle B$ and $\angle C = \angle A + \angle B$. Find $\angle C$.

- (A) 90° (B) 67.5°
(C) 45° (D) 112.5°

28. A triangle has sides 7 cm, 24 cm and 25 cm. Classified by its angles, it is:

- (A) Acute (B) Obtuse
(C) Equilateral (D) Right-angled

29. In triangle ABC, the exterior angle at A is 150° and the exterior angle at B is 140° . Find the interior angle at C.

- (A) 70° (B) 30°
(C) 40° (D) 110°

30. In triangle ABC, $AB = AC$. A point D lies on BC, and $AD = BD$. If $\angle B = 50^\circ$, find $\angle DAC$.

- (A) 50° (B) 80°
(C) 30° (D) 20°

31. Two sides of a triangle are 15 cm and 15 cm. The third side is a whole number that makes the triangle acute-angled. The smallest such whole-number third side is:

- (A) 2 cm
(C) 16 cm
- (B) 1 cm
(D) 22 cm

32. In triangle ABC, $\angle A = 100^\circ$ and $\angle B = \angle C$. The bisectors of $\angle B$ and $\angle C$ meet at I. Find $\angle BIC$.

- (A) 80°
(C) 140°
- (B) 40°
(D) 130°

33. A triangle has angles in the ratio 1 : 3 : 5. Side BC is opposite the SMALLEST angle. Which statement about the sides is correct?

- (A) BC is the longest side
(C) BC is the middle-length side
- (B) BC is the shortest side
(D) All three sides are equal

34. In triangle ABC, side BC is extended beyond C to D and beyond B to E. If the exterior angle $\angle ACD = 125^\circ$ and the exterior angle $\angle ABE = 118^\circ$, find $\angle A$.

- (A) 63°
(C) 55°
- (B) 117°
(D) 62°

35. An isosceles triangle has a perimeter of 40 cm. The base is 16 cm and each equal side is a whole number. The triangle is valid; what is each equal side?

- (A) 12 cm
(C) 8 cm
- (B) 16 cm
(D) 24 cm

36. In triangle ABC, $\angle A = 2\angle B$ and $\angle C = 3\angle B - 20^\circ$. Find $\angle B$.

- (A) 30°
(C) 25°
- (B) 40°
(D) $(200/6)^\circ \approx 33.3^\circ$

37. Two sides of a triangle are 30 cm and 18 cm. The third side x is a whole number with $18 \leq x \leq 30$. How many such values give a valid triangle?

- (A) 13
(C) 11
- (B) 12
(D) 48

38. In triangle ABC, $\angle B = 3\angle A$ and $\angle C = 5\angle A$. The triangle, by its angles, is:

- (A) Acute
(C) Obtuse
- (B) Right-angled
(D) Equiangular

39. In triangle ABC, the altitude from A meets line BC at H, and H lies on the extension of BC beyond C. Which interior angle of the triangle is obtuse?

- (A) The angle at C
(C) The angle at B
- (B) The angle at A
(D) No angle is obtuse

40. A triangle has two sides 21 cm and 13 cm. The third side is a whole number that makes the triangle isosceles. How many such whole-number values are there?

- (A) 1
(C) 0
- (B) 3
(D) 2

- 41.** In triangle ABC, $\angle A = 90^\circ$ and $\angle B = 30^\circ$. The altitude from the right angle A meets BC at H. In triangle ABH, what is $\angle BAH$?
- (A) 60° (B) 30°
(C) 90° (D) 45°
- 42.** Three sides of a triangle are $2k$, $5k$ and $6k$. Which classification by angles is forced for every $k > 0$?
- (A) Acute (B) Obtuse
(C) Right-angled (D) It depends on k
- 43.** In triangle ABC, $\angle A = 50^\circ$ and the bisector AD of $\angle A$ meets BC at D. In triangle ACD, $\angle ADC = 95^\circ$. Find $\angle C$.
- (A) 35° (B) 60°
(C) 25° (D) 85°
- 44.** A triangle has angles 36° , 36° and 108° . A second triangle has sides 9 cm, 9 cm and 9 cm. Which statement correctly classifies BOTH by sides and angles?
- (A) The first is scalene and obtuse; the second is equilateral and right
(B) The first is isosceles and acute; the second is isosceles and acute
(C) The first is equilateral and obtuse; the second is scalene and acute
(D) The first is isosceles and obtuse; the second is equilateral and acute
- 45.** In triangle ABC, $\angle A : \angle B : \angle C = 2 : 7 : 9$. An exterior angle is formed at the vertex with the SMALLEST interior angle. That exterior angle measures:
- (A) 160° (B) 20°
(C) 90° (D) 110°
- 46.** Two sides of a triangle are 19 cm and 6 cm. The third side is a whole number. The product of the smallest and largest possible third sides is:
- (A) 322 (B) 336
(C) 294 (D) 350
- 47.** In triangle ABC, $\angle B = \angle C$. The exterior angle at B (extending side CB beyond B) measures 128° . Find the vertex angle $\angle A$.
- (A) 52° (B) 128°
(C) 76° (D) 104°
- 48.** Four candidate side-triples (in cm) are: (5, 5, 9), (7, 2, 4), (10, 10, 19), (6, 8, 15). How many of these can form a valid triangle?
- (A) 1 (B) 3
(C) 2 (D) 4
- 49.** In triangle ABC, $\angle A = 84^\circ$. Points D and E lie on AB and AC so that $AD = AE$. In the small triangle ADE, what is the measure of $\angle ADE$?

- (A) 84° (B) 42°
(C) 96° (D) 48°

50. In triangle ABC, $\angle A = 40^\circ$, $\angle B = 60^\circ$ and $\angle C = 80^\circ$. The sides are a (opposite A), b (opposite B), c (opposite C). Which inequality chain is correct?

- (A) $c < b < a$ (B) $b < a < c$
(C) $a < b < c$ (D) $a < c < b$

51. In triangle ABC, side AB is extended beyond B to F. The exterior angle $\angle CBF = 116^\circ$, and inside the triangle $\angle A$ is 28° more than $\angle C$. Find $\angle C$.

- (A) 72° (B) 58°
(C) 44° (D) 64°

52. A triangle has sides that are three consecutive whole numbers and a perimeter of 21 cm. The triangle, classified by its angles, is:

- (A) Acute (B) Right-angled
(C) Obtuse (D) Equilateral

53. In triangle ABC, $\angle A = 70^\circ$. The bisectors of $\angle B$ and $\angle C$ meet at I, with $\angle BIC = 90^\circ + \frac{1}{2}\angle A$. By how many degrees does $\angle BIC$ exceed a right angle?

- (A) 70° (B) 55°
(C) 35° (D) 20°

54. Two sides of a triangle are 40 cm and 9 cm. The third side is a whole number. The number of whole-number values it can take is:

- (A) 18 (B) 16
(C) 17 (D) 31

55. In triangle ABC, $AB = AC$ and $\angle A = 36^\circ$. The bisector of base angle B meets AC at D. Find $\angle BDC$.

- (A) 36° (B) 72°
(C) 108° (D) 54°

56. In triangle ABC, the three interior angles are $(y + 10)^\circ$, $(2y - 30)^\circ$ and $(y + 20)^\circ$. Classified by its angles, the triangle is:

- (A) Right-angled (B) Obtuse
(C) Acute (D) Equiangular

57. An isosceles triangle has two equal sides each 13 cm and a base that is a whole number making the triangle obtuse at the apex. The number of such whole-number base lengths is:

- (A) 6 (B) 8
(C) 7 (D) 18

58. In triangle ABC, $\angle A = 2\angle B = 3\angle C$. Find the largest angle.

(A) 90°

(B) $(1080/11)^\circ \approx 98.2^\circ$

(C) 60°

(D) 108°

59. Two exterior angles of a triangle, at two different vertices, measure 117° and 121° . The exterior angle at the third vertex measures:

(A) 58°

(B) 238°

(C) 122°

(D) 120°

60. Two sides of a triangle are 14 cm and 5 cm, and the perimeter is a whole number of centimetres. The smallest possible whole-number perimeter is:

(A) 28 cm

(B) 19 cm

(C) 24 cm

(D) 29 cm

Solutions — Mathematics (Class 7), Chapter 7: A Tale of Three Intersecting Lines — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	A	31	B	41	A	51	C
2	D	12	B	22	D	32	C	42	B	52	A
3	D	13	A	23	B	33	B	43	B	53	C
4	C	14	B	24	B	34	A	44	D	54	C
5	B	15	B	25	D	35	A	45	A	55	B
6	B	16	D	26	C	36	D	46	B	56	C
7	A	17	A	27	A	37	A	47	C	57	C
8	D	18	A	28	D	38	C	48	C	58	B
9	D	19	B	29	D	39	A	49	D	59	C
10	A	20	D	30	C	40	D	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The angles $4k+5k+6k = 180^\circ$, so $k = 12^\circ$ and the angles are $48^\circ, 60^\circ, 72^\circ$. The interior angle at A is the smallest, 48° . The exterior angle at A is $180 - 48 = 132^\circ$. Reading the exterior angle as the interior 48° itself, or as 180 minus the largest $72^\circ = 108^\circ$, gives the wrong values.

2. (D) The third side x must satisfy $17 - 8 < x < 17 + 8$, i.e. $9 < x < 25$. The whole numbers strictly between are 10 through 24, which is $24 - 10 + 1 = 15$ values. Using $\leq$ at both ends gives $9..25 = 17$; using one inclusive end gives 16; subtracting $17 - 8 = 9$ mistakes the count for the difference.

3. (D) Sum: $(3x+5) + (2x-5) + 4x = 9x = 180^\circ$, so $x = 20^\circ$. Then $\angle A = 65^\circ, \angle B = 35^\circ, \angle C = 80^\circ$. The largest is $\angle C = 80^\circ$. Quoting $\angle A = 65^\circ$ as the largest, or stopping at $\angle B = 35^\circ$, are the traps.

4. (C) The exterior angle $\angle QRS$ equals the sum of the two remote interior angles $\angle P + \angle Q = 132^\circ$. With $\angle P = 2\angle Q$, we get $3\angle Q = 132^\circ$, so $\angle Q = 44^\circ$ and $\angle P = 88^\circ$. Quoting $\angle Q = 44^\circ$ as

$\angle P$, or using $180 - 132 = 48^\circ$ as the remote sum, gives the distractors.

5. (B) The bisector AD splits $\angle A$ into two equal parts, and $\angle BAD = 34^\circ$ is one half, so $\angle BAC = 2 \times 34^\circ = 68^\circ$. Quoting one half 34° as the whole, or taking $90 - 34 = 56^\circ$ as $\angle A$, are the named errors.

6. (B) With two sides 11 and 11, the third side x satisfies $0 < x < 22$. Whole numbers 1 through 21 give 21 values. Counting 0..22 inclusive gives 22; using $11 - 11 = 0$ as a usable side over-counts; excluding both ends of 1..21 wrongly gives 20.

7. (A) AM is a median, so it splits ABC into triangles AMB and AMC sharing point M on the straight line BC; $\angle AMB$ and $\angle AMC$ are supplementary, and $\angle AMB$ is the exterior angle of triangle AMC at M, equal to the sum of its two remote interiors $\angle MAC + \angle MCA$. A median need not be perpendicular, need not bisect the angle, and there is no reason $AM = AB$.

8. (D) Since $\angle B = \angle C$, the sides opposite them are equal; call each y . The side opposite $\angle A$ is 14. Perimeter: $14 + y + y = 50$, so $2y = 36$ and $y = 18$. The side opposite $\angle B$ is 18 cm. Quoting the leftover 36 as one side, or copying the given 14 cm, are the errors.

9. (D) The exterior angle at A equals $\angle B + \angle C$ and the exterior angle at B equals $\angle A + \angle C$, so their sum is $(\angle A + \angle B) + 2\angle C = (180 - \angle C) + 2\angle C = 180 + \angle C = 260$. Thus $5y + 4y = 260$, giving $9y = 260$ and $y = 260/9 \approx 28.9^\circ$. Treating the two exteriors as summing to 360 minus the third exterior, or rounding to a whole degree, give the decoys.

10. (A) The third angle is $180 - 42 - 96 = 42^\circ$, so two angles equal 42° and the triangle is isosceles. The longest side lies opposite the largest angle 96° , and the two sides opposite the equal 42° angles are equal and shorter. So the two 42° -opposite sides are equal and both shorter than the 96° -opposite side. Pairing the largest angle with a short side reverses the side-angle relationship.

11. (A) The exterior angle $\angle ACD$ equals the sum of the remote interior angles $\angle A + \angle B$. So $4\angle A = \angle A + 36^\circ$, giving $3\angle A = 36^\circ$ and $\angle A = 12^\circ$. Setting $4\angle A = 36$ gives 9° ; copying $\angle B = 36^\circ$; or reading the exterior 48° as the answer, are the traps.

12. (B) The third side x satisfies $16 - 9 < x < 16 + 9$, i.e. $7 < x < 25$, so x ranges over 8 to 24. The odd values are 9, 11, 13, 15, 17, 19, 21, 23 — that is 8 values. Counting all 17 integers, or miscounting the odd ones as 7, give the distractors.

13. (A) Let each base angle be b ; the two base angles total $2b$, and the vertex equals this sum, so vertex $= 2b$. Total: vertex + 2 base angles $= 2b + 2b = 180^\circ$, so $b = 45^\circ$ and vertex $= 2b = 90^\circ$. Taking $b = 45^\circ$ as the vertex, or solving $3b = 180 \rightarrow 60^\circ$, are the named errors.

14. (B) Let $\angle C = c$. Then $\angle B = c + 12$ and $\angle A = c + 36$. Sum: $3c + 48 = 180$, so $3c = 132$ and $c = 44^\circ$. Then $\angle B = 56^\circ$, $\angle A = 80^\circ$. The largest is $\angle A = 80^\circ$. Quoting $\angle B = 56^\circ$, or mis-

adding the offsets, give the wrong answers.

15. (B) An 8-8-8 triangle has three equal sides (equilateral) and three 60° angles (all acute). The 20-20-140 triangle has two equal angles, hence two equal sides (isosceles), and a 140° angle makes it obtuse. Calling the equilateral 'right', or the obtuse triangle 'acute', are the misconceptions.

16. (D) An interior angle and its exterior angle are supplementary: interior + exterior = 180° . With exterior = $3 \times$ interior, we get $4 \times$ interior = 180° , so interior = 45° . Solving $3 \times$ interior = $180 \rightarrow 60^\circ$, or quoting the exterior 135° , are the traps.

17. (A) The third side x satisfies $23 - 14 < x < 23 + 14$, i.e. $9 < x < 37$. Smallest whole value 10, largest 36. Difference = $36 - 10 = 26$. Using inclusive ends 9 and 37 gives 28; the difference $23 - 14 = 9$ is a decoy; the upper bound 37 is another.

18. (A) The exterior angle at A is 110° , so the interior vertex angle $\angle A = 180 - 110 = 70^\circ$. The two base angles are equal and sum to $180 - 70 = 110^\circ$, so each base angle = 55° . Quoting the apex 70° as a base angle, or copying 110° , are the named errors.

19. (B) $\angle BIC = 90^\circ + \frac{1}{2}\angle A = 90^\circ + \frac{1}{2}(80^\circ) = 90^\circ + 40^\circ = 130^\circ$. Using $180 - \angle A = 100^\circ$, the 100° -apex value 140° , or $\frac{1}{2}(180 - 80) = 50^\circ$, are the traps.

20. (D) Two equal sides (6, 6) make it isosceles. The angle opposite the longest side 10 is the largest; since $6^2 + 6^2 = 72 < 10^2 = 100$, that angle exceeds 90° , so the triangle is obtuse. Calling it acute ignores the wide angle opposite the long side; 'scalene' ignores the two equal sides.

21. (A) Sum: $(x-10) + (x+10) + 2x = 4x = 180$, so $x = 45$. The angles are $\angle A = 35^\circ$, $\angle B = 55^\circ$, $\angle C = 90^\circ$. A 90° angle makes the triangle right-angled. Mis-solving $4x = 180$ to a wrong x , or overlooking that $\angle C = 2x$ reaches exactly 90° , leads to the acute/obtuse traps.

22. (D) The tightest inequality has the two shortest summing past the longest: $(x - 2) + x > x + 2$, i.e. $2x - 2 > x + 2$, so $x > 4$. The smallest whole number greater than 4 is 5. Taking $x > 2$ from positivity gives 3; using $x \geq 4$ gives 4; over-shooting gives 6.

23. (B) The exterior angle $\angle ACD$ equals the sum of the two remote interior angles $\angle A + \angle B = 60 + 70 = 130^\circ$. (Equivalently $\angle C = 180 - 130 = 50^\circ$, and $180 - 50 = 130^\circ$.) Using $180 - 70 = 110$ reads the wrong vertex; quoting $\angle C = 50$; or $180 - 60 = 120$ picks the wrong remote angle.

24. (B) Let the vertex angle be v ; each base angle is $3v$. Sum: $v + 3v + 3v = 7v = 180^\circ$, so $v = 180/7 \approx 25.7^\circ$. Forgetting one base angle gives $5v = 180 \rightarrow 36^\circ$; treating base = vertex doubled gives others.

25. (D) Validity requires the third side $x < 12 + 12 = 24$, so $x \leq 23$. For the angle opposite x to be obtuse we need $x^2 > 12^2 + 12^2 = 288$, i.e. $x > 16.97$, so $x \geq 17$. Both conditions give $17 \leq x \leq 23$; the largest is 23 cm. 16 is just below the obtuse boundary; 24 violates the triangle inequality.

26. (C) $\angle AIB = 90^\circ + \frac{1}{2}\angle C = 90^\circ + \frac{1}{2}(90^\circ) = 90^\circ + 45^\circ = 135^\circ$. Using $\angle C$ itself 90° , using $\frac{1}{2}\angle C = 45^\circ$, or a mis-applied fraction giving 120° , are the traps.

27. (A) Since $\angle C = \angle A + \angle B$ and $\angle A + \angle B + \angle C = 180$, we get $2\angle C = 180$, so $\angle C = 90^\circ$. Solving $\angle A + \angle B = 90$ with $\angle A = 3\angle B$ gives $\angle B = 22.5^\circ$, $\angle A = 67.5^\circ$; quoting $\angle A$ as $\angle C$ is the trap.

28. (D) Check $7^2 + 24^2 = 49 + 576 = 625 = 25^2$. Since the square of the longest side equals the sum of the squares of the other two, the angle opposite 25 is exactly 90° , so the triangle is right-angled. Assuming the long side forces obtuse without checking, or calling a scalene triangle acute, are the errors.

29. (D) Interior $\angle A = 180 - 150 = 30^\circ$, interior $\angle B = 180 - 140 = 40^\circ$. Then $\angle C = 180 - 30 - 40 = 110^\circ$. Adding the exteriors $150 + 140 = 290$ then $360 - 290 = 70$ gives the exterior at C, not the interior; $\angle A$ and $\angle B$ mistaken as the answer give 30/40.

30. (C) Since $AB = AC$, $\angle C = \angle B = 50^\circ$, so $\angle BAC = 180 - 50 - 50 = 80^\circ$. In triangle ABD, $AD = BD$ makes it isosceles with $\angle BAD = \angle ABD = 50^\circ$. Then $\angle DAC = \angle BAC - \angle BAD = 80 - 50 = 30^\circ$. Quoting $\angle BAC = 80^\circ$, or $\angle BAD = 50^\circ$, or reading $50 - 30 = 20$ from a wrong subtraction, are the traps.

31. (B) With two equal sides 15 and 15, the apex angle opposite the base x is acute when $x^2 < 15^2 + 15^2 = 450$ ($x < 21.2$), and the two base angles, being equal and summing to less than 180 with the apex positive, stay below 90° for every positive x . So all whole bases from 1 up to 21 give an acute triangle; the smallest is 1 cm. Picking 2 over-restricts; 16 and 22 confuse the obtuse/validity boundaries.

32. (C) $\angle BIC = 90^\circ + \frac{1}{2}\angle A = 90^\circ + \frac{1}{2}(100^\circ) = 90^\circ + 50^\circ = 140^\circ$. Using $180 - 100 = 80^\circ$, using $\frac{1}{2}(180 - 100) = 40^\circ$, or the right-triangle value 130° , are the traps.

33. (B) The angles are 20° , 60° , 100° (since $9k = 180$, $k = 20$). BC is opposite the smallest angle 20° , and the shortest side lies opposite the smallest angle, so BC is the shortest. Pairing the smallest angle with the longest side reverses the relationship.

34. (A) Interior $\angle ACB = 180 - 125 = 55^\circ$ and interior $\angle ABC = 180 - 118 = 62^\circ$. Then $\angle A = 180 - 55 - 62 = 63^\circ$. Adding exteriors $125 + 118 = 243$ then $360 - 243 = 117$ gives the exterior at A; copying 55 or 62 mistakes a base angle for $\angle A$.

35. (A) Let each equal side be s ; $16 + 2s = 40$, so $2s = 24$ and $s = 12$ cm. Check validity: $12 + 12 = 24 > 16$, valid. Using $s = \text{base} = 16$, taking half the base 8, or quoting the leftover 24

for two sides as one side, are the errors.

36. (D) $\angle A + \angle B + \angle C = 2\angle B + \angle B + (3\angle B - 20) = 6\angle B - 20 = 180$, so $6\angle B = 200$ and $\angle B = 200/6 \approx 33.3^\circ$. Treating $6\angle B = 180 \rightarrow 30^\circ$, or $5\angle B = 200 \rightarrow 40^\circ$ by dropping a coefficient, are the named slips.

37. (A) The overall valid range is $30 - 18 < x < 30 + 18$, i.e. $12 < x < 48$, satisfied by every x with $18 \leq x \leq 30$. So all integers 18 through 30 qualify: $30 - 18 + 1 = 13$ values. Dropping one endpoint gives 12; both gives 11; the full range width is a decoy.

38. (C) $\angle A + 3\angle A + 5\angle A = 9\angle A = 180$, so $\angle A = 20^\circ$, $\angle B = 60^\circ$, $\angle C = 100^\circ$. Since $\angle C = 100^\circ > 90^\circ$, the triangle is obtuse. Mis-solving $8\angle A = 180$ to land $\angle C$ at 90° gives the right-angled trap.

39. (A) When the foot of the altitude from A falls outside segment BC, beyond C, the angle at C is obtuse — the perpendicular overshoots past the obtuse vertex C. Attributing the obtuseness to A, or to the far vertex B, mistakes which vertex the foot landed beyond.

40. (D) For isosceles, the third side must equal 21 or 13. A third side of 21 (giving 21, 21, 13) is valid since $21 + 13 > 21$. A third side of 13 (giving 21, 13, 13) needs $13 + 13 = 26 > 21$, valid. Both work, so 2 values. Counting only one, or wrongly rejecting 13, gives the other options.

41. (A) In triangle ABH, $\angle AHB = 90^\circ$ (AH is perpendicular to BC) and $\angle B = 30^\circ$, so $\angle BAH = 180 - 90 - 30 = 60^\circ$. Copying $\angle B = 30^\circ$, the right angle 90° , or assuming an isosceles 45° split, are the named errors.

42. (B) Validity: $2k + 5k = 7k > 6k$, valid for all $k > 0$. The largest side is $6k$; compare $(6k)^2 = 36k^2$ with $(2k)^2 + (5k)^2 = 4k^2 + 25k^2 = 29k^2$. Since $36k^2 > 29k^2$, the angle opposite $6k$ exceeds 90° , so the triangle is obtuse for every k . The fixed ratio means the type cannot depend on k .

43. (B) AD bisects $\angle A = 50^\circ$, so $\angle DAC = 25^\circ$. In triangle ACD: $\angle C = 180 - \angle ADC - \angle DAC = 180 - 95 - 25 = 60^\circ$. Using the full $\angle A = 50^\circ$ instead of its half gives 35° ; quoting the half 25° as $\angle C$; or $180 - 95 = 85^\circ$ forgetting $\angle DAC$, are the traps.

44. (D) Two equal angles ($36^\circ, 36^\circ$) mean two equal sides, so the first is isosceles, and its 108° angle makes it obtuse. The 9-9-9 triangle has three equal sides (equilateral) and three 60° angles (acute). Calling the first scalene ignores the equal angles; calling the equilateral 'right' is wrong since all its angles are 60° .

45. (A) $2k + 7k + 9k = 18k = 180$, so $k = 10^\circ$ and the angles are $20^\circ, 70^\circ, 90^\circ$. The smallest interior angle is 20° (at A). The exterior angle there is $180 - 20 = 160^\circ$. Quoting the interior 20° ; using $180 - 90 = 90$ at the wrong vertex; or $180 - 70 = 110$, are the traps.

46. (B) Third side x satisfies $19 - 6 < x < 19 + 6$, i.e. $13 < x < 25$. Smallest whole value 14, largest 24. Product = $14 \times 24 = 336$. Using $14 \times 23 = 322$, $14 \times 21 = 294$, or $14 \times 25 = 350$, are the named slips.

47. (C) The exterior angle at B is 128° , so interior $\angle B = 180 - 128 = 52^\circ$. Since $\angle B = \angle C$, $\angle C = 52^\circ$ too. Then $\angle A = 180 - 52 - 52 = 76^\circ$. Quoting $\angle B = 52^\circ$ as $\angle A$, copying 128° , or $2 \times 52 = 104^\circ$ as $\angle A$, give the distractors.

48. (C) (5,5,9): $5+5 = 10 > 9$ valid. (7,2,4): $2+4 = 6 < 7$ invalid. (10,10,19): $10+10 = 20 > 19$ valid. (6,8,15): $6+8 = 14 < 15$ invalid. So 2 are valid. Miscounting (6,8,15) as valid because $8+15 > 6$ gives 3; rejecting (10,10,19) gives 1.

49. (D) Triangle ADE is isosceles with $AD = AE$, so its base angles $\angle ADE = \angle AED$ are equal and sum to $180 - \angle A = 180 - 84 = 96^\circ$, giving each = 48° . Quoting $\angle A = 84^\circ$ for the base angle, taking half of $84 = 42^\circ$, or using the full 96° as one base angle, are the traps.

50. (C) Larger angles face longer sides. Since $\angle A = 40^\circ < \angle B = 60^\circ < \angle C = 80^\circ$, the opposite sides satisfy $a < b < c$. Reversing the relationship, or swapping the middle terms, gives the wrong chains.

51. (C) The exterior angle $\angle CBF$ equals the sum of the remote interiors $\angle A + \angle C = 116^\circ$. With $\angle A = \angle C + 28$, we get $(\angle C + 28) + \angle C = 116$, so $2\angle C = 88$ and $\angle C = 44^\circ$. Quoting $\angle A = 72^\circ$ as $\angle C$, splitting 116 evenly to 58° , or copying 64° from a wrong route, are the traps.

52. (A) Consecutive whole numbers summing to 21: 6, 7, 8. Check the largest side 8: $6^2 + 7^2 = 36 + 49 = 85 > 8^2 = 64$, so the angle opposite 8 is below 90° ; the other angles are smaller still, so all are acute. Assuming the longest side forces obtuse, or expecting a 6-8-10 right triangle, are the errors.

53. (C) $\angle BIC = 90^\circ + \frac{1}{2}\angle A = 90^\circ + 35^\circ = 125^\circ$. The amount by which it exceeds a right angle (90°) is $\frac{1}{2}\angle A = 35^\circ$. Quoting $\angle A = 70^\circ$ itself, using $90 - 35 = 55^\circ$, or a wrong fraction giving 20° , are the distractors.

54. (C) Third side x satisfies $40 - 9 < x < 40 + 9$, i.e. $31 < x < 49$. Whole numbers 32 through 48 give $48 - 32 + 1 = 17$ values. Using one inclusive end gives 18; using the difference 31 is a decoy; mis-counting gives 16.

55. (B) Base angles: $\angle B = \angle C = (180 - 36)/2 = 72^\circ$. The bisector of $\angle B$ gives $\angle DBC = 36^\circ$. In triangle BDC: $\angle BDC = 180 - \angle DBC - \angle C = 180 - 36 - 72 = 72^\circ$. Using $180 - 72 = 108^\circ$, taking half of $72 = 36^\circ$ as $\angle BDC$, or $90 - 36 = 54^\circ$, are the misconception traps.

56. (C) Sum: $(y+10) + (2y-30) + (y+20) = 4y = 180$, so $y = 45$. The angles are 55° , 60° , 65° , all below 90° , so the triangle is acute. Mis-solving $4y = 180$ to a wrong y , or assuming the spread forces a right or obtuse angle, are the errors.

57. (C) Base b satisfies validity $b < 13 + 13 = 26$, so $b \leq 25$. For the apex (opposite the base) to be obtuse: $b^2 > 13^2 + 13^2 = 338$, so $b > 18.38$, i.e. $b \geq 19$. Thus b ranges 19 through 25, giving $25 - 19 + 1 = 7$ values. Starting at 18 gives 8; miscounting gives 6; the bound 18 is a decoy.

58. (B) Let $\angle A = t$. Then $2\angle B = t$ gives $\angle B = t/2$, and $3\angle C = t$ gives $\angle C = t/3$. Sum: $t + t/2 + t/3 = 11t/6 = 180$, so $t = 1080/11 \approx 98.2^\circ$, the largest angle ($\angle A$). Assuming a neat 90° or 60° by mis-balancing the ratios, or dropping a term to get $t = 108^\circ$, are the traps.

59. (C) The three exterior angles (one per vertex) of any triangle sum to 360° . So the third is $360 - 117 - 121 = 122^\circ$. Finding the third interior angle and stopping gives 58° ; adding the two given exteriors gives 238° ; assuming all equal gives 120° .

60. (D) The third side x satisfies $14 - 5 < x < 14 + 5$, i.e. $9 < x < 19$, so the smallest whole third side is 10. The smallest perimeter is $14 + 5 + 10 = 29$ cm. Using $x = 9$ (the boundary, which fails the strict inequality) gives 28; quoting the third-side bound 19; or using $x = 5$ gives 24.